Assignment -1

Soft Computing Technique (NAI-602)

1. Explain the **architecture of neural networks** with suitable diagrams.
2. Differentiate between **single layer feed-forward network** and **multilayer feed-forward network**.
3. Explain **recurrent neural networks** and their applications.
4. Describe various **learning techniques in neural networks**.
5. Explain **auto-associative memory** and **hetero-associative memory** with examples.
6. Compare **biological neurons and artificial neurons**.
7. Explain the **working of a perceptron** with example of training for AND logic
8. Explain the **architecture of Back Propagation Network** with diagram.
9. Describe the **Back Propagation learning algorithm step by step**.
10. Explain the **training process of multilayer perceptron networks**.
11. Discuss the **factors affecting back propagation training**.
12. Explain how **single layer neural networks are solved using perceptron learning**.
13. There is three-layered multi-layer perceptron in fig. apply backpropagation algorithm

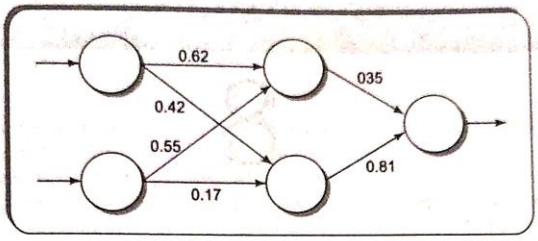

Figure 7.13 Three-layered Multi-layer Perceptron

And let the patterns to be learned be:

Input1	*Input2*	*Target*
0	1	0
1	1	1

Apply backpropagation algorithm.

Note: Last date of Submission 14/03/2026